

Tiny 2-Leg Spinal CPG — Network Reduction

Change log: what was modified, why, and what it produced. Target: preserve flexor/extensor alternation while cutting neurons and synapses. Final operating point **k = 0.25, m = 0.40** (25% connectivity, 40% neurons, ~4% of baseline synapse count, since synapses scale as m^2k).

1. Starting problem

Reducing connectivity degraded muscle alternation progressively: $\text{corr}(\text{Force-E}, \text{Force-F})$ fell from **-0.953** at $k=1.00$ to **-0.895** at $k=0.50$ and **-0.665** at $k=0.25$. Earlier variants (Q, T, V) had tried to fix this by retuning the flexor activation-gate threshold alone, without success (-0.79 to -0.88).

2. Main changes, in order

#	Change	Why	Result
1	Inhibitory weight compensation New --inh-comp, set to 3.0. Scales W_{RG2INE} and $W_{INE2RGF}$ (36→108, -24→-72).	Diagnostic plots showed RG-F flat at ~145 Hz (10 Hz p-p) while RG-E swung 50-265 Hz. $\text{corr}(\text{RG})$ still read -0.92 because correlation is amplitude-blind. Cause: $P_{RG_RECIP_E} = 0.15k = 0.0375$, a 4x loss of inhibitory synapses. Fix conserves total drive by holding p-w constant as count falls.	$\text{corr}(\text{RG})$ -0.927 → -0.962 (baseline -0.964). $\text{corr}(\text{F})$ unchanged at -0.69: necessary but not sufficient.
2	Extensor gate recalibration Exposed $X0_E / K_E$; set $X0_E$ 0.6→1.10, K_E 8→15.	The decisive fix. $X0_E=0.6$ was calibrated when RG-E troughed at 0 Hz. After reduction RG-E troughs at ~50 Hz, i.e. $r/100 = 0.5$, sitting <i>below</i> the threshold, so the E gate never closed (gate range 0.71, Force-E floor 11.9). Every prior variant tuned the F gate and left $X0_E$ at its baseline-era value.	$\text{corr}(\text{F})$ -0.69 → -0.963 . Gate range 0.71 → 1.00; Force-E floor 11.9 → 1.3.
3	Flexor gate $X0_F = 1.42$, $K_F = 30$.	$X0_F$ must sit inside RG-F's actual 120-155 Hz band. Values above its 99th percentile (1.50-1.55) left the gate shut (range 0.31, Force-F capped at 4.9). Steepness beyond ~40 squared off the waveform and cost 0.16 of $\text{corr}(\text{F})$.	Completed the recipe. 3-seed spread 0.002 at the time.
4	NEST kernel RNG seeded <code>rng_seed</code> added to <code>SetKernelStatus</code> ; runs use --threads 1.	Only numpy was seeded, so --seed never controlled connectivity. Identical config and seed gave -0.955 and -0.847 in different sweeps. A determinism test then showed threads=1 bit-identical across repeats, threads=8 and 64 not (spread 0.042-0.046): NEST is non-reproducible at these population sizes with many threads.	Runs now exactly reproducible. Revealed that comparisons finer than ~0.08 in earlier sweeps had been noise.

3. Changes tested and rejected

Tested	Outcome
RG-F drive raised (I_E_RGF 0.9→1.5, $W_{RG_REC_F}$ 5.5→7.5)	Flexor peak rose 15.1→16.4 but its floor rose 3.0→5.5 and $\text{corr}(\text{F})$ fell. Band translated upward, not widened.
RG-F drive lowered (I_E_RGF 0.9→0.1, $W_{RG_REC_F}$ →4.5)	RG-F span stayed 21-26 Hz across the full 15x drive range. Its dynamic range is structurally fixed by connectivity, not by drive.
Deeper reduction: $k=0.15/m=0.40$, $k=0.25/m=0.30$	Both viable with gates recentred (corr -0.94). Not adopted: floors worse and no seed confirmation completed.
$k=0.15$ with $m=0.30$	Reduction limit. $\text{corr}(\text{RG})$ itself collapses to -0.73/-0.78 while gates stay healthy: the rhythm generator fails, not the readout.
Finer rate sampling (rate-update 100→20 ms)	Reverted. The rate feeds the gate and Ia loop, so 20 ms bins let spike-count noise drive the dynamics: $\text{corr}(\text{RG})$ -0.91→-0.55, gate saturated. Smoothing is now applied at plot time only.

4. Final configuration

Group	Values
Scale	$k = 0.25$, $m = 0.40$ — per leg: RG-E/F 40 each, InE/InF 20 each, motor / muscle / Ia 40 each
Probabilities	$P_{RG_RECIP_E}$ 0.0375, $P_{RG_RECIP_F}$ 0.075, P_{RG_REC} 0.03, P_{M2MUS} 0.20, P_{IA2IN} 0.0625, P_{MOTOR_RECIP} 0.0625, P_{COMM_F} 0.055, P_{COMM_E} 0.025
Inhibition	<code>inh_comp</code> 3.0 (W_{RG2INE} 108, $W_{INE2RGF}$ -72); <code>inh_comp_f</code> 1.0 (W_{RG2INF} 18, $W_{INF2RGE}$ -48, <i>uncompensated</i>)

Group	Values
Activation gate	X0_E 1.10, K_E 15, X0_F 1.42, K_F 30
Unchanged	I_E_RGF 0.9, W_RG_REC_F 5.5, all cell parameters, delays, Ia gains, brainstem drive, gait pacing; STDP off
Simulation	30 s, resolution 0.2 ms, chunk 100 ms, rate-update 100 ms, threads 1 , paced gait 520 ms, stance 0.5

5. Results (8 seeds)

Condition	corr(F)	Force-E floor	Force-F floor	Force-F peak
Baseline k=1.00	-0.953	~1.1	~1.1	~16.9
k=0.25 / m=0.40, fixed gate	-0.911 ± 0.073	1.40	1.41	~13
same, per-network gate	-0.969 ± 0.005	3.07	4.18	~15.2
best seed (52345)	-0.973	1.5	1.8	13.9

Per-network gate calibration (thresholds set from each network's own measured RG percentiles) removes the seed-to-seed spread but raises both force floors. Neither variant matches baseline on correlation *and* floor depth simultaneously; both should be reported.

6. Interpretation and open items

The mechanism is not preserved. At k=0.25 RG-F no longer bursts: it is tonically active at ~140 Hz with a ~24 Hz ripple, while RG-E swings over 210 Hz. Alternation is recovered by compensated inhibition plus a rescaled readout, not by a restored half-centre. Note the distinction between the two kinds of change made: inhibitory compensation is a circuit-level correction for lost synapses, whereas the gate parameters are a readout recalibration that leaves the network untouched.

The flexor fails first. Every failure mode traced to RG-F; RG-E was robust throughout.

Open: (i) the flexor floor remains above baseline on most seeds; (ii) inh_comp_f was never swept — the F→E pathway is the one uncompensated connection; (iii) measured cycle frequency reads 1.667 Hz against a 1.923 Hz pacing driver, unresolved as either an estimator artefact at 100 ms bins or a real period difference; (iv) whether per-network gate calibration is acceptable depends on the claim being made — standard for "alternation is preserved", a caveat for "drop-in replacement".